Servicenow

Session Agenda

1.Session Agenda:

- UI Policy task
- Data Policy task
- Difference between UI Policy and Data Policy
- Same rule using Dictionary
- Difference between Dictionary vs Policies
- Notification task (send email when record created)
- Import data from Excel using Data Source
- Transform Map task
- ACL
- Practices

Task 1 - UI Policy

- On Asset Request table, create a UI Policy.
- Condition: Asset Type = Laptop
- Action: Make Justification mandatory and show Warranty Period field.
- If Asset Type is not Laptop, Justification becomes optional and Warranty Period is hidden.
- Test the behavior directly from the form.

Task 2 - Data Policy

- Create a Data Policy on Asset Request table.
- Condition: Request Status = Approved
- Action: Assigned To and Approval Date must be mandatory.
- Enable 'Apply to import sets' so the rule also works during imports.
- Try saving a record without these fields to verify the validation.

Difference Between UI Policy and Data Policy

- UI Policy works only on the form in the browser (client side).
- UI Policy can hide, show, read-only, or mandatory fields while user edits the form.

- Data Policy works on the server side and applies everywhere.
- Data Policy mainly enforces data validation even if data comes from import, API, or script.

Task 3 - Same Requirement Using Dictionary

- Open the dictionary entry for field: Justification.
- Set Mandatory = true from Dictionary instead of UI Policy.
- Compare the result with the UI Policy version.
- Notice that Dictionary makes the field mandatory all the time, not only when Asset Type = Laptop.

Dictionary vs Policy

- Dictionary is static: always applied for every record.
- UI Policy and Data Policy are dynamic: applied only when condition is true.
- Dictionary is best for permanent rules such as always mandatory or default value.
- Policies are better when the rule depends on another field or condition.

Task 4 - Notification / Email

- Create an email notification for Asset Requests.
- When Request Status changes to Approved, send an email to the Requested By user.
- Email subject: Your asset request has been approved.
- Email body should include Request Number, Asset Name, and Assigned To.
- Test by approving a sample request and checking the email preview.

Task 5 - Import Data from Excel

- Prepare an Excel sheet with columns: Asset Name, Asset Type, Requested By, Status.
- Go to System Import Sets > Load Data.
- Upload the Excel file as a Data Source.
- Create an Import Set Table and map the fields to Asset Request table.
- Run Transform Map and verify that records are created correctly.

ACL Tasks – Asset Request System

- Practice Access Control (ACL) in ServiceNow based on Asset Request system.

Task 1 – Create Roles

- Create roles:
- asset_viewer
- asset_editor
- asset_admin
- Assign them to users.

- Create + Write + Read
- Read
- Read+ Write
- Read+ Delete
- All

Task 2 – Read ACL

- Create Read ACL on Asset Request table
- Only asset_viewer can read records.

Task 3 – Write/Delete ACL

- Only asset_editor can edit and delete records.
- Test with different users.

Task 4 – Field ACL

- Restrict 'Cost' field
- Only asset_admin can see it.

Task 5 – Condition ACL

- User can access record only if Assigned To = current user.

Task 6 – Script ACL

- Use script:
- answer = current.age < 5;
- Allow access based on condition.

Task 7 – Comparison

- ACL = Security
- UI Policy = UI behavior
- Data Policy = Data validation
- Dictionary = Field setup

Practice With students:

- Import an Excel file where one approved record is missing Assigned To.
- Because of the Data Policy, the record should fail during import.
- Then fix the Excel file and import again successfully.
- Finally approve another request and confirm that the email notification is sent.

Practices

Task 1 — UI Policy (Restaurant)

Create UI Policy:
Table: u_restaurant_order
Condition: Dish Name = Pizza
Actions:
  Make Quantity → Mandatory
  Make Table Number → Visible

Test:
When selecting Pizza → fields change instantly

Task 2 — Data Policy

Create Data Policy:
Table: u_restaurant_order
Rules:
  Dish Name → Mandatory
  Apply to all (check Apply to all)

Test:
Try insert via script/import → should still enforce

Task 3 — Dictionary Enhancement

Open Dictionary:
Field: u_dish_name
Set:
  Mandatory = true
  Max length = 100
  Default value = Pasta

Compare:
Dictionary = permanent
UI Policy = dynamic
Data Policy = server-side

Task 4 — Notification (Email)

Create Notification:
Table: u_restaurant_order
Trigger: Insert
Send to: Assigned Waiter

Example message:
New order assigned to you

Task 5 — Import Data (Excel)

Upload Excel with columns:
- Dish Name
- Quantity
- Table Number

Steps:
System Import Sets → Data Sources
Upload file

Task 6 — Transform Map

Map fields:

Dish Name → u_dish_name
Quantity → u_quantity
Table Number → u_table_number

Run transform
Verify records created

Task 7 — Basic ACL (Read Access)

Create ACL:
Type: record
Table: u_restaurant_order
Operation: read
Condition:
  Only allow users with role: restaurant_user

Task 8 — Role-Based ACL

Create roles:
- chef
- manager
- waiter

ACL rules:
Chef → Read only
Waiter → Create + Read
Manager → Full access

Task 9 — Field-Level ACL

Restrict Total Amount field
Only Manager can see it

ACL:
Type: field
Field: u_total_amount
Role: manager

Task 10 — Conditional ACL

Allow Waiter to see only their orders

Result:
Waiter sees only his orders

Task 11 — Write Restriction ACL

Prevent editing after order is completed

Condition:
State = Closed → No write

Task 12 — Create ACL

Allow only Waiter to create orders

Operation: create
Role: waiter

Task 13 — Delete ACL

Only Manager can delete orders

Operation: delete
Role: manager

Task 14 — Advanced ACL (Script + Role)

Allow Chef to update only Quantity

Type: field
Field: u_quantity
Operation: write
Role: chef
Script: check current.u_dish_name.changesTo() is false

Homework session#4

UI Policy Actions (1) | UI Policy Related List Actions

for text | Search | Actions on selected rows... | New

UI policy = If type IPD

Field name	Mandatory	Visible	Read only
u_admitted_date	True	True	Leave alone

Table: Patient [u_patient]

Application: Global

Active: ☑

* Short description: If type IPD

When to Apply

Conditions: Add Filter Condition | Add OR Clause

Type | is | IPD | AND

Type: General

Gender: -- None --

Type: IPD

nder: -- None --

* Admitted Date:

Task 1: UI Policy

If Patient Type = IPD → Admitted Date visible and mandatory

1. Patient Table → Configure → UI Policies
2. Create new policy
3. Set condition and actions

List Control
New record View: List Controls

Table Patient [u_patient]

Label

Display Configuration | Edit configuration | Performance

List edit insert row ☑

List edit type Save immediately (cell edit mode)

Submit

Data Policy
Phone Is mandatory

Description

Conditions Add Filter Condition Add OR Clause

-- choose field -- -- oper --

Update Delete

Related Links

Convert this to UI Policy

[SN Utils] Versions (1)

Data Policy Rules for text Search

Data Policy = Phone Is mandatory

Field name	Mandatory
u_phone	True

Patients Number Search Actions

Data Policy Exception: The following fields are mandatory: Phone

Number	Address	Admitted Date	Age	Blood Group	Gender	Patient ID	Patient N
PAT0001001	test	2026-04-04	23	A	Female	231	Patient 1
PAT0001002	test	2026-04-09	22	B	Male	2323	patient 2
(empty)							

Insert a new row...

Task 2: Data Policy

Phone Number must be mandatory always

1 System Definition → Data Policies

2 Create new policy

Notification [sysevent_email_action]
Notify Doctor New App

When to send | Who will receive | What it will contain

Notifications can be sent (if the specified **Conditions** are met) un

- A record is **Inserted** or **Updated** into the **Table** specified ab
- The specified event is fired
- Via a Flow Action

Send when: Record inserted or updated

Inserted: ☑

Conditions: Add Filter Condition | Add OR Clause

-- choose field --

Notification
Notify Doctor New App View: Advanced

Preview Notification | Up

Verdana | 8pt

Hi Dr:/${u_doctor.u_doctor_name}

New App Created: ${sys_created_on}

Patient Name: ${u_patient.u_patient_name}

Time: ${u_time}

Appointment Number: ${u_appointment_number}

Date: ${u_date}

Fields
- Appointment Number
- Created
- Created by
- Date
- Department
- Doctor
- Number
- Patient
- Status
- Sys ID
- Tags
- Time
- Updated
- Updated by
- Updates

Users

Users/Groups in fields

Available: Department, Doctor ID, Doctor Name, Number, Phone, Specialization, Tags, Updated

Selected: Doctor.Email

Groups

Subscribable ☐

Created	Recipients	Subject	Type	Notification type	User ID
trying from servicenow	test@test.com	New Appoinment	send-ready	SMTP	(empty)

Task 3: Notification

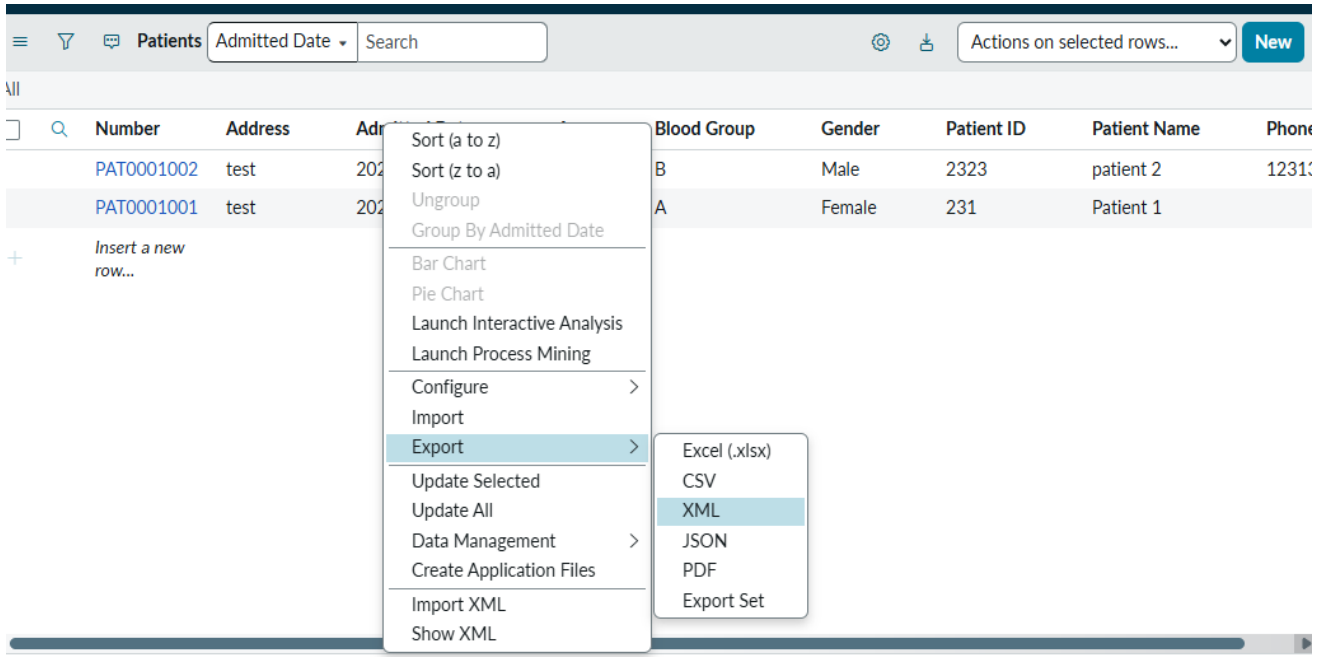
Notify doctor when appointment created

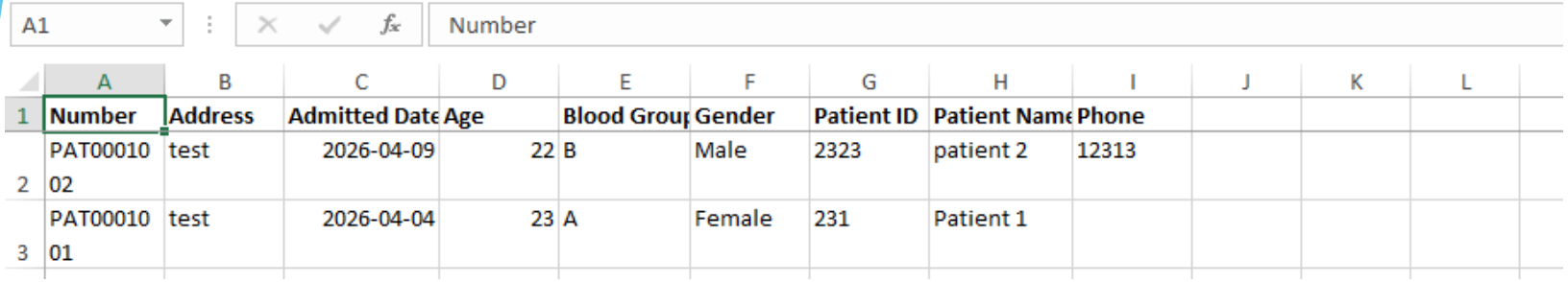
1. System Notification → Notifications
2. Create new notification

Task 4: Excel Import

Import patient data from Excel

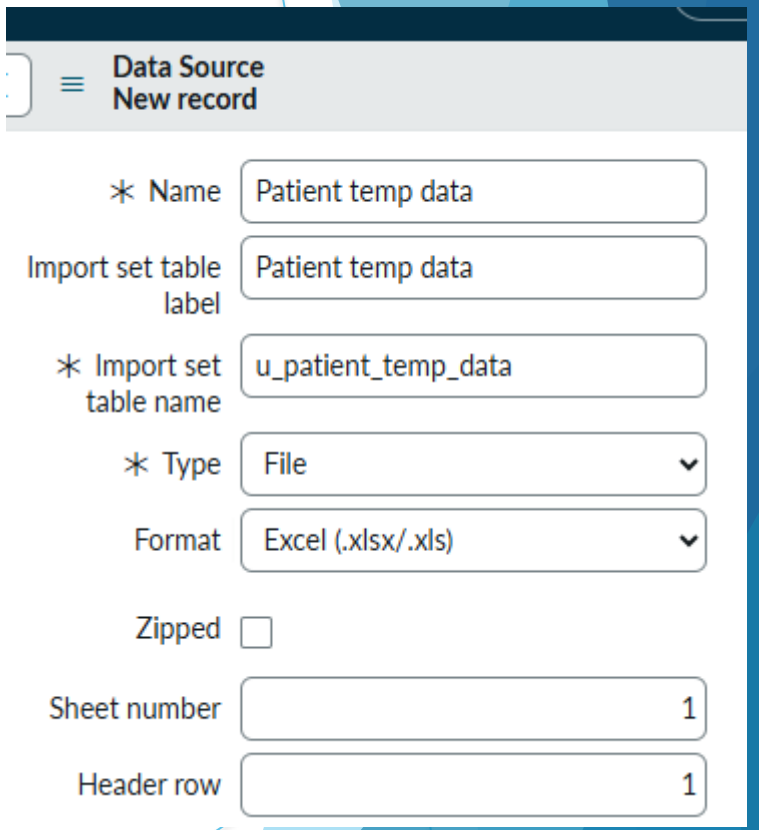
1. System Import Sets → Load Data
2. Upload file
3. Create transform map





A1 | Number

	A	B	C	D	E	F	G	H	I
1	Number	Address	Admitted Date	Age	Blood Grou	Gender	Patient ID	Patient Name	Phone
2	PAT0001002	test	2026-04-09	22	B	Male	2323	patient 2	12313
3	PAT0001001	test	2026-04-04	23	A	Female	231	Patient 1	



Task 4: Excel Import

Import patient data from Excel

1. System Import Sets → Load Data
2. Upload file
3. Create transform map

Update set: Default [Glo...

Table Transform Map
New record

* Name	from temp to per table
* Source table	u_patient_temp_data
Active	☑
Run business rules	☑
Enforce mandatory fields	No
Copy empty fields	☐
Create new record on empty coalesce fields	☐
Application	Global
Created	
* Target table	Patient [u_patient]
Order	100
Run script	☐

Test loading 20 rows, please review the list below - Return to data source

Progress

Name	ImportProcessor
State	Running
Message	Creating table u_patient_temp_data

Refreshing...

Favorites History Admin Table Transform Map - from temp t

Table Transform Map
from temp to per table

Transform
Index Coalesce Fields
Run Point Scan
[SN Utils] Versions (1)

Field Maps (9) Transform Scripts

Source field	Target field
u_blood_group	u_blood_group
u_admitted_date	u_admitted_date
u_address	u_address
u_phone	u_phone
u_age	u_age
u_gender	u_gender
u_patient_id	u_patient_id
u_patient_name	u_patient_name
u_number	u_number

Progress

Name	ImportProcessor
State	Complete
Completion code	Success
Message	Processed: 2, inserts 2, updates 0, errors 0, empty and ignored 0, ignored

Favorites History Workspaces Admin Patients

Application scope: Global
Update set: Default [Glo...

Patients Admitted Date Search Actions on selected rows... New

Number	Address	Admitted Date	Age	Blood Group	Gender	Patient ID	Patient Name	Phon
PAT0001002	test	2026-04-09	22	B	Male	2323	patient 2	1231
PAT0001002	test	2026-04-09	22	B	Male	2323	patient 2	1231
PAT0001001	test	2026-04-04	23	A	Female	231	Patient 1	

Questions About (UI Policy –Data Policy- Notifications –Import Set)

Which of the following field attributes can be controlled by a UI Policy? (Choose three.)

A. Hidden
B. Mandatory
C. Read-only
D. Necessary
E. Write-only

Answer: A,B,C

- UI Policies manage Visible (Hidden), Mandatory, and Read-only states. There is no "Necessary" or "Write-only" state.

Which statement best describes the difference between a UI Policy and a Data Policy?

A. UI Policies run on the server, while Data Policies run in the browser.
B. Data Policies apply to all data entering the system (Import Sets, Web Services), while UI Policies only apply to data entered on a form through a browser.
C. UI Policies can be used to make fields mandatory, but Data Policies cannot.
D. Data Policies can include complex client-side scripts, whereas UI Policies cannot.

Answer: B

- Data Policies are server-side and enforce rules for all data entry points (Web Services, Import Sets). UI Policies are client-side and only affect browser form interactions.

Can a UI Policy be converted into a Data Policy?

A. Yes, but only if it does not contain scripts or visibility actions and is marked as Global.

B. No, Data Policies must always be created from scratch.

C. Yes, all UI Policies can be converted regardless of their configuration.

D. No, only Data Policies can be converted into UI Policies

Answer: A

•Conversion is possible if the UI Policy is Global, has no scripts, and no visibility actions.

What are the three essential parts of a Notification?

A. Trigger, Condition, Result

B. Why to send, How to send, When to delete

C. Table, Field, Script

D. When to send, Who will receive, What it will say

Answer: D

- These are the three core tabs/questions asked when configuring any ServiceNow notification.

An administrator wants to use the same content for different notification channels (e.g., Email and SMS). Which feature should they use?

A. Common notification content
B. Default notification content
C. Related notification content
D. Actionable notification content

Answer: A

•Common notification content allows administrators to reuse the same message body across multiple delivery channels.

In the Import Set process, what is the purpose of a Transform Map?

A. To physically move the data from the source file to the staging table.

B. To determine the relationship between fields in an Import Set table and fields in an existing ServiceNow table.

C. To verify the credentials of the user performing the import.

D. To automatically delete duplicate records in the source file.

Answer: B

- The Transform Map provides a guide for moving data from the "Staging" (temporary) table to the "Target" (actual) table.

During an import, which field property is used to check if a record already exists in the target table to prevent duplicates?

A. Mandatory
B. Coalesce
C. Unique Key
D. Primary Value

Answer: B

- **Coalesce** fields act as a unique key during import; if a match is found, the existing record is updated instead of creating a duplicate.

At what stage of the import process is data actually written to the final target table?

A. Load Data
B. Create Staging Table
C. Run Transform
D. Inspect Discovery

Answer: C

- **Data is only moved from the staging table to the destination table when you Run Transform.**

Which role is required to create or update Access Control (ACL) rules?

A. user_admin
B. security_admin
C. admin
D. personalize_control

Answer: B

Only the security_admin role can manage ACLs. Even standard admins must **elevate their privileges** to this role to make changes.

https://www.youtube.com/watch?v=FsOx8PAbVBk

What is the correct execution order when ServiceNow evaluates an ACL?

A. Role, Script, Condition
B. Condition, Role, Script
C. Role, Condition, Script
D. Script, Condition, Role

Answer: C

ServiceNow evaluates ACLs in this specific order: it first checks if the user has the required **Roles**, then evaluates the **Conditions** (if any), and finally runs the **Script**. If any step fails, access is denied.

Which of the following is the highest level of security in ServiceNow?

A. Business Rules
B. UI Policies
C. Client Scripts
D. Access Control Lists (ACLs)

Answer: D

ACLs are server-side security rules, making them the most secure. UI Policies and Client Scripts are client-side and can be bypassed by advanced users.

In what order does the system search for a match when evaluating a record-level ACL?

A. Table level, then Field level
B. Field level, then Table level
C. Parent table, then Child table
D. Scripted rules, then Role rules

Answer: A

The system first checks for a match at the table (row) level. If access is not granted to the record, the user cannot see any fields, regardless of field-level ACLs.

What does the "table.none" notation represent in an ACL?

A. Access to all fields in the table
B. Access to the entire record (row-level access)
C. Access to a specific field
D. No access is granted

Answer: B

"Table.none" applies to the **record itself**. "Table.*" applies to all **fields** in that record where no specific field ACL exists.

If a user has three different roles and an ACL requires only one of those roles to pass, what will be the result?

A. Access Denied
B. Access Granted
C. The system checks the next ACL
D. Access is restricted to Read-only

Answer: B

ServiceNow uses a "logical OR" for roles within a single ACL; if a user matches **at least one** of the required roles, they pass that specific check

Homework session#5

Thank you